CASEY CRANE

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EDUCATION

BS 2015, MS 2020, PhD 2026, Mathematics, University of Texas at Dallas

PUBLICATIONS

Dissertation

- "Consensus-breaking global Hopf bifurcation in memory-based multi-agent systems." PhD Dissertation. Advisor: Dr. Wieslaw Krawcewicz.

Journal Publications

- Chaoquan Chen, Casey Crane, and Travis Hensley. "Global Hopf bifurcation in symmetric configurations of distributed delay differential equations" *Communications in Pure and Applied Analysis* 27 (2026): 170–194.
- Ziad Ghanem, Casey Crane, Jingzhou Liu. "Global bifurcation of non-radial solutions for symmetric sub-linear elliptic systems on the planar unit disc." *Journal of Fixed Point Theory and Applications* 26 (2024): 44.
- Yameng Duan, Casey Crane, Wieslaw Krawcewicz, and Huafeng Xiao. "Periodic solutions in reversible symmetric second order systems with multiple distributed delays" *Journal of Differential Equations* 401 (2024): 282–307.

Preprints

- Casey Crane. "Global Hopf bifurcation and symmetric periodic solutions in multi-agent systems with neutral distributed delays" arXiv:2604.20740 (2026). In revision at SIADS.
- Casey Crane. "Trust as hysteresis: analytically tractable strategies for the iterated prisoner's dilemma" Preprint (2026). Submitted to J. Theor. Biol.
- Casey Crane and Ziad Ghanem. "Unbounded branches of non-radial solutions to semilinear elliptic systems on a disc and their patterns." arXiv:2504.06519 (2025). In revision at TMNA.
- Casey Crane and Ziad Ghanem. "Unbounded branches of non-radial solutions to semilinear elliptic systems on a unit ball in $\mathbb{R}^3$ and their patterns" arXiv:2512.14957 (2025). Preprint.

RESEARCH SUMMARY

My research uses equivariant degree theory to study symmetry-breaking bifurcation and global continuation in dynamical systems, with a focus on delay equations and highly symmetric networks of interacting agents. Building on the study of consensus protocols in multi-agent systems, I ask what happens after consensus breaks down: I use topological methods to prove the existence and global continuation of branches of symmetric periodic solutions, and to classify them by their spatio-temporal symmetry. These are periodic multiconsensus states: the agent population fragments into clusters that oscillate out of phase. This work also led me to identify a class of delay equations I call pseudoneutral: equations written in the implicit form of neutral equations that are in fact of retarded type, and that arise naturally as the mean-field limit of populations with distributed neutral delays. I pair these abstract results with targeted numerical analysis to determine stability and period, and I develop applications in UAV formation control and in networks of coupled financial markets. More broadly, I am interested in how memory shapes behavior in systems of interacting agents—in multi-agent networks, and in analytically tractable models of trust in strategic interaction.

Teaching Experience

Teaching Associate, University of Texas at Dallas *Spring 2026*
Instructor of record: Differential Calculus.

Teaching Assistant, University of Texas at Dallas *Fall 2021 – Fall 2025*
Courses supported: Calculus I–III (single variable, vector, and multivariable calculus), Differential Equations, Functional Analysis (graduate).

Industry Experience

Systems and Processes Engineer / Lead Data Analyst, Good Faith Energy, Addison, TX *2020–2024*
(Full-time through 2022, part-time thereafter.) Designed and built company-wide automated data-analysis, quoting, and engineering calculation tools, including an NEC-compliant automated wire-sizing application for the electrical engineering team; performed statistical modeling of labor costs and job costing.

Technical Skills

Python (NumPy/SciPy, JiTCDDE), Mathematica, GAP, LaTeX

Talks

- *Invited.* "Global symmetric Hopf bifurcation using equivariant degree." Hunan Normal University, Changsha, China, 2023.
- *Contributed.* "Global Hopf bifurcation in symmetric configurations of distributed delay differential equations." SIAM Texas-Louisiana Section Annual Meeting, Baylor University, 2024.